

Title: The Impact of Remote Working on Cybersecurity Risks and Defence Strategies in Fintech Organisations: A Proposed Study on SOC Effectiveness.

Module: Contemporary Issues in Computer Science.

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An AI Use Declaration is included in Appendix B

1. Title of Research

The Impact of Remote Working on Cybersecurity Risks and Defence Strategies in Fintech Organisations: A Proposed Study on SOC Effectiveness

2. Introduction and Background

Financial technology has become crucial to modern finance, with fintech businesses supplying services through digital banking, digital payments, mobile applications and online financial infrastructures. The sector is also becoming increasingly shaped by advanced technologies, with the AI in fintech market forecast to grow from \$22.5 billion in 2023 to \$79.4 billion by 2030 (Statista, 2024). As shown in Figure 1, the AI in fintech market is forecast to grow substantially by 2030, reinforcing the need for stronger cybersecurity and SOC capability in technology-driven financial services. Cele and Kwenda (2024) show that digital banking adoption is affected by cybersecurity threats such as identity theft, malware, phishing and vishing, demonstrating that trust is essential to digital financial services. Akinyemi (2024) similarly argues that fintech and digital payment systems improve accessibility and efficiency, but expose organisations to fraud, data breaches, malware, AI-enabled threats and regulatory pressure. This supports the concept that fintech cybersecurity is not merely a technological issue, but also a trust, governance and resilience concern.

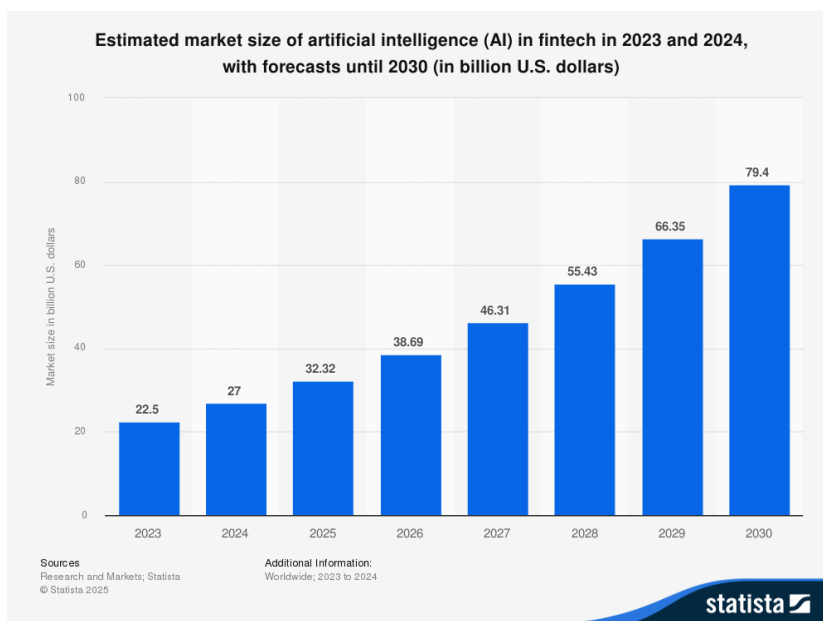


Figure 1: Estimated market size of AI in fintech from 2023 to 2030. Source: Statista (2025).

Remote and hybrid working have further changed this risk environment. Odeh and Odeh (2025) argue that remote financial workspaces face persistent threats including phishing, ransomware and credential theft, while basic controls such as MFA and VPNs are common but advanced approaches such as Zero Trust remain unevenly adopted. Onyemere et al. (2024) also show that remote working in banking is associated with cybersecurity risks, supervision challenges, technological constraints and collaboration issues. This is relevant because fintech staff may access client data, payment systems and cloud services from scattered locations.

In response, fintech organisations deploy defence strategies such as MFA, VPNs, Zero Trust, SIEM, endpoint monitoring, encryption and regulatory compliance. However, AlBenJasim et al. (2023) argue that fintech cybersecurity frameworks remain uneven and that existing regulations may not fully address sector-specific risks. This links with Assessment 1 sources, which found that defence strategies are often discussed separately and their effectiveness depends on implementation, integration and human behaviour (Carjuman, Fook and Hlaing, 2025; Madat, Poriya and Srivaramangai, 2025). Therefore, this position paper suggests a document-analysis study analysing how remote working affects financial cybersecurity threats, defence measures and SOC efficacy. This establishes a firm framework for the proposed study effort.

3. Research Question and Objectives

Primary Research Question:

How does existing literature and industry guidance explain the impact of remote working on cybersecurity risks, defence strategies and SOC effectiveness in fintech organisations?

Objectives:

1. To analyse existing literature on cybersecurity risks introduced by remote working in fintech organisations.

2. To evaluate how defence strategies such as MFA, VPNs, Zero Trust, SIEM and endpoint monitoring are presented as mitigating remote-working risks in fintech organisations.
3. To identify gaps in current research, with specific attention to SOC effectiveness in remote and hybrid fintech environments.

4. Brief Justification of Proposed Research

This research is justified on academic, practical and regulatory grounds. Academically, it expands Assessment 1 by addressing a gap between fintech cybersecurity risk, remote-working defence techniques and SOC effectiveness. Existing studies examine fintech threats and controls, including phishing, ransomware, data breaches, MFA, VPNs and SIEM, but often treat these as separate issues rather than as an integrated operational system (Kamuangu, 2024; Umoga et al., 2024; Alsumayt et al., 2025; Qollakaj and Larsson, 2023; Khan et al., 2024). Jones et al. (2023) show that work-from-home increased SOC responsibility for endpoint monitoring, privacy and burnout, while Vielberth et al. (2020) argue that SOC research remains fragmented. Jukanti (2025) also links remote work to inadequate policies, unsecured networks and training gaps.

Practically, the need for this research is reinforced by industry statistics showing that cyber incidents in the financial industry increased from 856 incidents in 2013 to 3,336 incidents in 2024, while incidents involving data disclosure reached 927 in 2024 (Petrosyan, 2024). This suggests that financial organisations continue to suffer a high level of cyber exposure despite the increased adoption of cybersecurity measures. As shown in Figure 2, cyber events in the financial industry remained high in 2023 and 2024, reinforcing the need to review whether defence measures and SOC capabilities are successful in practice. Fintech organisations require evidence that security tools boost operational outcomes, not simply that they are employed. Sohal (2025) links SOC maturity to stronger incident response performance, while Tariq et al. (2025) show that alert fatigue can reduce analyst effectiveness. Ezaz and Khan (2024)

demonstrate the value of SIEM and machine-learning integration for incident response, although the SCADA focus limits direct fintech application.

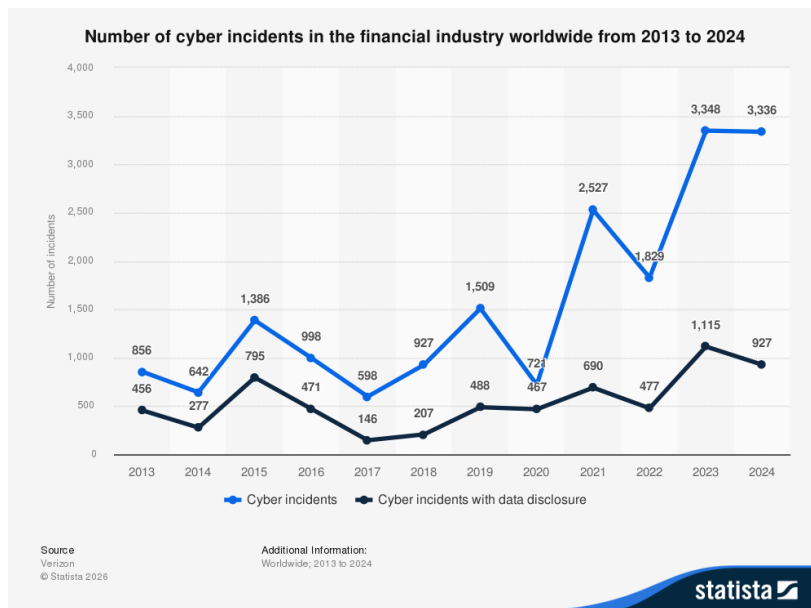


Figure 2: Number of cyber incidents in the financial industry worldwide from 2013 to 2024. Source: Statista (2026).

Regulatory justification is also strong. DORA strengthens ICT risk management, incident reporting, testing and third-party oversight for financial entities (Clausmeier, 2022), while Janciute (2025) argues that financial cybersecurity requires precautionary planning for emerging risks. A DORA-focused ICT risk study also highlights banking-sector exposure and resilience requirements. AlBenJasim et al. (2024) and Lubis et al. (2026) further show that fintech and non-bank financial organisations need sector-specific frameworks and SOC effectiveness measures. As a result, this document-analysis study would combine academic, industrial, and regulatory evidence to promote more evidence-based SOC resilience in remote fintech environments. It would also help clarify where current evidence is strong, where it remains vendor-driven, and where future fintech cybersecurity research should focus for practical SOC governance decisions today.

5. Motivation

My motivation for this research comes from an academic and professional interest in SOC and Blue Team cybersecurity. Assessment 1 showed that defence strategies

such as MFA, VPNs, SIEM and Zero Trust are important, but their value depends on implementation, monitoring and everyday use (Carjuman, Fook and Hlaing, 2025; Madat, Poriya and Srivaramangai, 2025). This is significant because rules that appear successful in principle may behave differently when remote workers use home networks, personal devices, and cloud services.

The SOC-specific gap is especially important. Alexander (2025) argues that threat detection and response depend on the combination of people, processes and technology, while Revaclier Romain (2021) shows that SOC capability needs structured maturity evaluation. Woods and Seymour (2024) also argue that cybersecurity control effectiveness should be evidence-based rather than assumed. This supports my focus on whether remote-working defence strategies genuinely support SOC visibility, alert triage and response.

The professional motivation is also linked to my aim of entering SOC and Blue Team work. Sources on SOC analyst roles and next-generation SOC resilience show that analysts need technical judgement, human awareness and adaptive response capability (Cooper, 2025; Sugumar, 2026). Therefore, this proposal is academically relevant, realistic and connected to my career direction within fintech security practice.

6. Positions: Main Arguments in the Literature

The literature can be divided into three linked sections that illustrate how remote working affects fintech cybersecurity risks, defence strategies and SOC effectiveness.

The first position is that remote and hybrid working increase fintech cyber risk by widening access beyond controlled office environments. Assessment 1 sources showed that remote workers face phishing, insecure home networks, BYOD risks and weaker supervision (Klint, 2023; Scott, 2022; Chigada and Daniels, 2021; Rah, 2023). Further research into more recent literature supports this view. Nizamuddin (2025) synthesises remote-work evidence and identifies human factors, insecure Wi-Fi, VPN misuse, device misuse, policy non-compliance and weak organisational communication as recurring vulnerabilities. Kamuangu (2024) similarly identifies data breaches, phishing, malware, regulatory compliance and MFA as core fintech concerns, while Ngcai and Gundu (2025) argue that BYOD improves flexibility in

financial organisations but creates data protection, compliance and monitoring challenges. Atencio (2025) adds that remote work and IoT widen attack surfaces through home networks, personal devices and cloud collaboration tools. As a result, remote working does not eliminate all fintech risks but rather increases them by distributing sensitive work across more devices, networks, and locations.

The second position is that defence strategies are necessary, but not automatically effective. MFA, VPNs, Zero Trust, SIEM and endpoint monitoring are common controls, yet their value depends on implementation quality, integration and user behaviour. Maheswara and Chennuri (2024) argue that adaptive MFA can combine behavioural biometrics, device context and risk scoring, but Assessment 1 showed that MFA can still be weakened by phishing, SIM swapping and session hijacking (Carjuman, Fook and Hlaing, 2025). VPNs show the same issue: Thomas (2023) links VPNs to teleworking productivity and security, but VPNs cannot solve endpoint compromise or weak governance alone. Zero Trust literature strengthens this critique. Tsai, Lee and Shieh (2024) argue that remote work exposes perimeter-security weaknesses, while Mavroudis (2024) frames ZTNA around “never trust, always verify”. Hassan, Hassan and Khan (2024) apply this to fintech cloud infrastructures, where continuous authentication and microsegmentation reduce unauthorised access. Atencio (2025) supports integrated defence, reporting strong results for Zero Trust and AI-enhanced XDR, but this also confirms that isolated controls are insufficient. Axon et al. (2021) and Woods and Seymour (2023) reinforce that control effectiveness depends on context and evidence, not merely deployment.

The third position, which is adopted in this work, is that SOC efficacy is under-researched in comparison to risks and tools. SOC turn controls into operational outcomes through log collection, alert triage, investigation and response. González-Granadillo, Zarzosa and Diaz (2021) describe SIEM as central to modern SOC visibility, while Onwubiko and Ouazzane (2019) argue that effective SOC also require service onboarding, business alignment and continuous analyst monitoring. Recent SOC research deepens this gap. Baruwat Chhetri et al. (2024), Turcotte, Labrèche and Paquette (2025), and Kearney et al. (2023) show that alert fatigue, false positives and limited business context can reduce analyst effectiveness. Federated learning, SOCaaS and AI-assisted SOC research suggest improvements in collaboration,

detection and triage, but much of this work is not fintech-specific (Khramtsova et al., 2020; Arfaoui, Mrabet and Jemai, 2023; Antoniou, 2025).

Overall, the literature indicates that remote working raises financial risk, technical controls are necessary but implementation-dependent, and SOC efficacy remains a significant evidence gap. Existing studies often imply that deploying MFA, VPNs, SIEM or Zero Trust improves security, but fewer examine whether these tools improve SOC visibility, alert triage and incident response in remote fintech environments. This proposed study therefore argues for document analysis that compares academic, industry and regulatory evidence to clarify where remote-working defence strategies support SOC outcomes and where current knowledge remains assumed, fragmented or weak within fintech security practice today.

7. Research strategy

This proposed study would use qualitative document analysis to examine existing academic literature, industry reports, regulatory guidance and cybersecurity frameworks on remote working, fintech cybersecurity, defence strategies and SOC effectiveness. Document analysis is suitable because it allows pre-existing texts to be selected, appraised and synthesised systematically, rather than requiring interviews or surveys with SOC professionals (Bowen, 2009; Morgan, 2022). This is suited for fintech since SOC operations may contain sensitive information, such as detection rules, incident response procedures, tool configurations, and internal vulnerabilities. Using public documents therefore reduces access barriers and ethical risks while still allowing a critical investigation of the research question.

The method would follow a transparent process of finding, selecting, organising, evaluating and interpreting documents. Bowen (2009) defines document analysis as a systematic procedure for reviewing printed and electronic material, while Kayesa and Shung-King (2021) argue that document analysis is strongest when the method is clearly aligned to the research question, search strategy and contribution of documents to findings. Morgan (2022) also notes that document analysis can support research where fieldwork is impractical but warns that researchers must justify document selection carefully. This study would therefore record search terms,

databases, inclusion criteria, exclusion criteria and reasons for rejecting sources. Documents would be collected from three source categories (See in appendix A).

The selected documents would be analysed using reflexive thematic analysis. Braun and Clarke (2021, 2024) emphasise that thematic analysis requires methodological coherence, reflexivity and transparency rather than simply listing topics. Naeem et al. (2023) provide a structured process for moving from keywords and codes to themes and interpretation, which would help organise the analysis. The coding would be guided by the research question but remain open to unexpected patterns. Anticipated themes include remote working and expanded attack surface, BYOD and endpoint security, phishing and human factors, VPN limitations, Zero Trust access control, SIEM visibility, alert triage, SOC maturity, incident response and regulatory resilience.

To improve quality, the study would compare evidence from academic, industry, and regulatory documents. This triangulation is important because document analysis can be weakened by poor source selection, unclear criteria or over-reliance on one type of evidence (Chanda, 2021; Sankofa, 2022). A SPIDER-style eligibility logic would help clarify the sample, phenomenon of interest, design, evaluation and research type (Amir-Behghadami, 2021). Reliability would be supported through a documented audit trail, consistent criteria and transparent coding decisions. Validity would be supported by linking each theme directly to evidence and acknowledging limitations. The study would not claim to measure live SOC performance; instead, it would synthesise what existing documents reveal about remote-working defence strategies and SOC effectiveness in fintech. This fits the proposal scope while remaining feasible, ethical and connected to Assessment 1 findings and future practice. As this study uses qualitative document analysis and does not involve human participants, a recruitment strategy is not applicable.

8. Ethical Considerations and Limitations

This proposed study has lower ethical risk than interview-based research because it uses publicly available documents rather than human participants, personal data or confidential organisational material. Therefore, informed consent, withdrawal rights and interview confidentiality would not apply. However, cybersecurity research still requires ethical care because documents may discuss vulnerabilities, attack

techniques, weak controls and sensitive organisational practices. Reidsma, van Ham and Continella (2023) show that cybersecurity research can create privacy, service-disruption and misuse risks, while Shou (2012) highlights long-standing debates around disclosure, human subjects and data sharing. Ababneh et al. (2025) further argue that cybersecurity must be guided by accountability, transparency, fairness and proportionality. Therefore, the study would avoid reproducing exploit instructions and would focus on implications for SOC practice.

A second ethical issue is evidence quality. Vendor or industry documents may overstate the effectiveness of tools such as MFA, AI monitoring or endpoint platforms. Ahi and Valizadeh (2025) show that AI is dual-use, supporting both defensive detection and malicious automation, while Pomerleau and Maimon (2022) argue that cybersecurity practice should be evidence-based rather than assumption-led. Dalal (2025) and Nee et al. (2026) also show that cybersecurity policies and platform implementation require context-aware governance, not simple tool deployment.

The main limitation is that document analysis cannot capture direct SOC analyst experience or prove live fintech SOC performance. Public documents may omit internal alert fatigue, escalation delays or informal workarounds. Moilanen et al. (2022) warn that documents are not always produced for research purposes, while Holst et al. (2024) show that document-based monitoring needs clear indicators, reliability and comparability. Reliability and validity would be supported through transparent inclusion criteria, comparison of academic, regulatory and industry sources, an audit trail, critical evaluation of claims, and direct links between themes and source evidence throughout the final written research report.

9. Conclusion

This position paper has argued that remote working has intensified cybersecurity risks in fintech organisations by expanding access across home networks, personal devices, cloud services and distributed working practices. The growth of AI-driven fintech and the high number of financial-sector cyber incidents reinforce the urgency of this issue. Although defence strategies such as MFA, VPNs, Zero Trust, SIEM and endpoint monitoring are discussed, the report shows that their value depends on implementation quality, integration, user behaviour and SOC capability rather than

simple deployment. The proposed study responds to this gap through qualitative document analysis, synthesising academic, industry and regulatory evidence on remote-working risks, defence strategies and SOC effectiveness. This approach is realistic because fintech SOC operations may involve confidential procedures that are difficult to access through primary research. Overall, the study would support more evidence-based SOC resilience in fintech environments where digital trust, operational continuity and regulatory compliance are essential.

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Appendices

Appendix A

Academic sources would include peer-reviewed articles from Google Scholar, Scopus and OneSearch on remote working, fintech cybersecurity, MFA, VPNs, Zero Trust, SIEM, SOC maturity and incident response. Industry sources would include cybersecurity reports and white papers, used critically because vendor documents may promote specific tools. Regulatory sources would include FCA operational resilience guidance, DORA materials and cybersecurity frameworks. Sources would be included if they address remote or hybrid working, fintech or financial services, defence strategies, SOC monitoring, incident response or cyber resilience. Sources would be excluded if they are unrelated to cybersecurity, purely opinion-based, outdated for technical topics, or unsupported by evidence.

Appendix B

ChatGPT was used as an academic support tool during the preparation of this assessment. It was used to support brainstorming, planning, explanation of concepts, report structure, search phrase development, proofreading and presentation preparation.

All AI-assisted content was reviewed, edited and adapted by me. The final topic, research question, objectives, argument, source selection, critical evaluation and submitted work reflect my own understanding and academic judgement. AI was not used to fabricate sources, create false references, collect primary data or replace my own analysis.